

Tax Subsidy Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Subsidy Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.47 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 41 (36) bps/month with a t-statistic of 3.98 (3.66), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap) is 34 bps/month with a t-statistic of 3.56.

1 Introduction

Financial markets efficiently incorporate information into asset prices when investors correctly assess the risk exposures of different securities. Understanding how firm characteristics relate to systematic consumption risk remains a central challenge in asset pricing, particularly in identifying which observable signals capture cross-sectional variation in expected returns through their connection to investors' marginal utility of consumption. We identify Tax Subsidy Intensity (TSI) as a novel predictor of stock returns that operates through firms' differential exposure to aggregate consumption risk. Firms with varying degrees of tax subsidization exhibit distinct sensitivities to macroeconomic shocks that affect households' intertemporal consumption decisions, creating systematic return patterns that standard factor models fail to explain.

Tax Subsidy Intensity captures firms' exposure to consumption risk through multiple economic channels grounded in consumption-based asset pricing theory. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with higher tax subsidies face greater sensitivity to persistent shocks in aggregate consumption growth, as government support mechanisms amplify their exposure to fiscal policy uncertainty that directly affects households' consumption-savings decisions. These subsidized firms exhibit heightened sensitivity to time-varying risk aversion, consistent with the habit formation models of [Campbell and Cochrane \(1999\)](#), where investors demand higher compensation for holding assets whose returns covary more strongly with consumption during periods of elevated marginal utility.

The state-dependent nature of tax subsidies creates asymmetric exposure to consumption disasters, aligning with the rare disaster framework of [Barro \(2006\)](#) and [Gabaix \(2012\)](#). During economic downturns when marginal utility is high, heavily subsidized firms experience sharper valuation adjustments as government support becomes uncertain and fiscal constraints bind, while less subsidized firms provide

relative consumption hedging benefits. This mechanism generates cross-sectional variation in disaster risk loadings that manifest as persistent return differentials, with investors requiring higher expected returns for firms whose tax benefits amplify their sensitivity to tail consumption events.

Tax subsidization further influences firms' covariance with the stochastic discount factor through its impact on operating leverage and cash flow duration, consistent with [Zhang \(2005\)](#) and [Lettau and Wachter \(2007\)](#). Firms receiving substantial tax subsidies exhibit higher effective operating leverage as these benefits represent quasi-fixed claims on future cash flows that increase sensitivity to aggregate productivity shocks affecting consumption growth. The resulting variation in cash flow elasticity to consumption shocks creates systematic differences in risk premia, where high-TSI firms command higher expected returns due to their stronger positive covariance with the intertemporal marginal rate of substitution during periods when consumption growth is particularly valuable to investors.

We find strong evidence that Tax Subsidy Intensity predicts cross-sectional variation in expected returns. A value-weighted long-short strategy that buys high-TSI stocks and sells low-TSI stocks earned an average monthly return of 36 basis points with a t-statistic of 3.65 over our sample period from June 1963 to June 2024. The strategy achieved an annualized Sharpe ratio of 0.47, placing it in the 87th percentile among 212 documented anomalies in the factor zoo. These results demonstrate economically significant compensation for bearing consumption risk exposure captured by tax subsidy variation.

The TSI strategy's alpha remained robust across all standard factor models, ranging from 30 to 44 basis points per month with t-statistics exceeding 2.93. Most notably, the strategy delivered a monthly alpha of 41 basis points (t-statistic = 3.98) relative to the Fama-French six-factor model, indicating that existing factors fail to capture the consumption risk channel through which tax subsidies affect returns.

The strategy’s significant negative loading of -0.28 on the RMW profitability factor (t-statistic = -5.80) suggests that high-TSI firms’ returns move opposite to profitability shocks, consistent with their heightened sensitivity to consumption risk during economic distress.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the TSI strategy maintained an alpha of 34 basis points per month with a t-statistic of 3.56. The strategy’s performance persisted across size quintiles, generating a significant monthly alpha of 29 basis points (t-statistic = 2.45) among the largest stocks where implementation is most feasible. After accounting for transaction costs using bid-ask spreads, the net Sharpe ratio of 0.42 ranked in the 97th percentile of the anomaly distribution, confirming that TSI captures genuine variation in consumption risk premia rather than market frictions.

Our study advances the consumption-based asset pricing literature by identifying tax subsidies as a measurable firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Lettau and Wachter \(2007\)](#) and [Parker and Julliard \(2005\)](#) establish theoretical links between firm characteristics and consumption risk, we provide direct empirical evidence that tax subsidy intensity serves as an observable proxy for firms’ covariance with marginal utility growth. Unlike [Jagannathan and Wang \(2007\)](#) who focus on aggregate consumption measures, we demonstrate that firm-level tax characteristics contain information about idiosyncratic consumption risk exposures that generate significant return premia even after controlling for standard risk factors and related accounting-based anomalies.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that TSI represents a distinct source of systematic risk rather than a repackaging of known anomalies. Our finding that TSI maintains significant predictive power after controlling for closely related signals including book-to-market ([Fama and French \(1993\)](#)), earnings-to-price ([Basu \(1977\)](#)), and

R&D intensity ([Chan et al. \(2001\)](#)) from the Journal of Finance and Journal of Financial Economics demonstrates that tax subsidies capture a unique dimension of consumption risk. The strategy’s robust performance net of transaction costs, with generalized alphas remaining significant across all factor model specifications, distinguishes our results from many anomalies that disappear after accounting for implementation frictions as shown by [Novy-Marx and Velikov \(2016\)](#).

Our findings have broader implications for understanding the role of government policy in asset pricing through consumption-based channels. The systematic return patterns associated with tax subsidy intensity suggest that fiscal policy interventions create cross-sectional variation in firms’ exposure to aggregate consumption risk, providing a micro-foundation for how government actions affect equilibrium risk premia. This evidence supports consumption-based models where time-varying disaster risk and state-dependent marginal utility drive expected returns, while offering practitioners a implementable strategy for capturing these risk premia. The persistence of the TSI premium across different market conditions and portfolio construction methods indicates that markets efficiently price consumption risk exposure, but this pricing reflects rational compensation for systematic risk rather than behavioral biases or limits to arbitrage.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over our sample period, which spans several decades of annual observations. COMPUSTAT’s standardized reporting ensures consistency in variable definitions across firms and time periods, making it well-suited for constructing our tax subsidy intensity measure. signal relies on two

key variables from firms’ financial statements. The investment tax credit income account (ITCI) represents the amount of investment tax credits recognized in a firm’s income statement during the fiscal year. These credits arise from government tax incentive programs designed to encourage capital investment in specific assets or activities. The second variable, market equity at the COMPUSTAT data date ($me_{atadate}$), captures the market value of common equity calculated as the product of shares outstanding on the date when COMPUSTAT records the observation. We construct the Tax Subsidy Intensity signal by dividing the tax subsidy received by a firm by its market value. This ratio captures the magnitude of tax subsidies received by a firm relative to its market value and requires non-missing values for both components to ensure data quality. The resulting signal identifies firms that have received tax subsidies or those particularly effective at utilizing available tax incentive programs.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal’s interquartile range spans -0.00 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 4.07% of CRSP names, which on average make up 5.06% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.36% per month with a t-statistic of 3.65. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.30% to 0.44% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.28, with a t-statistic of -5.80 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 398 stocks and an average market capitalization of at least \$865 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 3.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TSI strategy achieves an average return of 29 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 22 to 38 bps/month

with t-statistics between 1.80 and 3.16.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy’s gross (net) Sharpe ratio of 0.47 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TSI strategy (red line).² Ignoring trading costs, a \$1 invested in the TSI strategy would have yielded \$10.05 which ranks the TSI strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TSI strategy would have yielded \$7.65 which ranks the TSI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 68 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 208 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 1.32, with the minimum t-statistic occurring when controlling for RD over market cap. Controlling for all six closely related anomalies, the t-statistic on TSI is 1.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 30-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.94, which is achieved when controlling for RD over market cap. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.56.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the TSI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes TSI grows to \$3136.63.

8 Conclusion

This study provides compelling evidence that Tax Subsidy Intensity (TSI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.47 gross (0.42 net), indicating strong economic viability even after accounting for transaction costs. Most notably, the strat-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

egy’s ability to generate a statistically significant alpha of 34 basis points per month (t-statistic = 3.56) after controlling for the Fama-French five factors, momentum, and six closely related strategies underscores that TSI captures unique information about expected returns not subsumed by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TSI emerges as a meaningful addition to the factor zoo, offering insights into how tax-related corporate characteristics influence asset prices. For practitioners, the strategy’s net Sharpe ratio of 0.42 suggests that TSI can be profitably implemented in real-world portfolios, particularly given its orthogonality to traditional value, profitability, and cash flow-based strategies. The persistence of abnormal returns after controlling for related strategies such as book-to-market, earnings-to-price, and cash flow measures indicates that TSI captures a distinct dimension of firm value that reflects the market’s assessment of firms’ tax planning efficiency and its implications for future cash flows.

Despite these robust findings, several limitations warrant consideration. First, while our analysis controls for transaction costs, the implementation of TSI strategies may face additional real-world frictions such as short-sale constraints and capital requirements that could affect realized returns. Second, the economic mechanism underlying the TSI anomaly requires further investigation—whether it reflects risk compensation, mispricing, or limits to arbitrage remains an open question. Third, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets requires additional research.

Future research should explore several promising avenues. First, investigating the time-varying nature of TSI returns and their relationship with macroeconomic conditions, particularly tax policy changes, could provide insights into the economic drivers of the anomaly. Second, examining the interaction between TSI and corporate governance characteristics may reveal whether tax planning effectiveness varies

systematically with management quality or board oversight. Third, extending the analysis to include the post-2017 Tax Cuts and Jobs Act period would help assess the strategy's robustness to major tax regime changes. Finally, developing a theoretical asset pricing model that explicitly incorporates tax subsidies as a state variable could provide a more fundamental understanding of why TSI predicts returns. Overall, this study establishes TSI as a valuable addition to the empirical asset pricing literature and opens new directions for understanding the role of corporate tax strategies in financial markets.

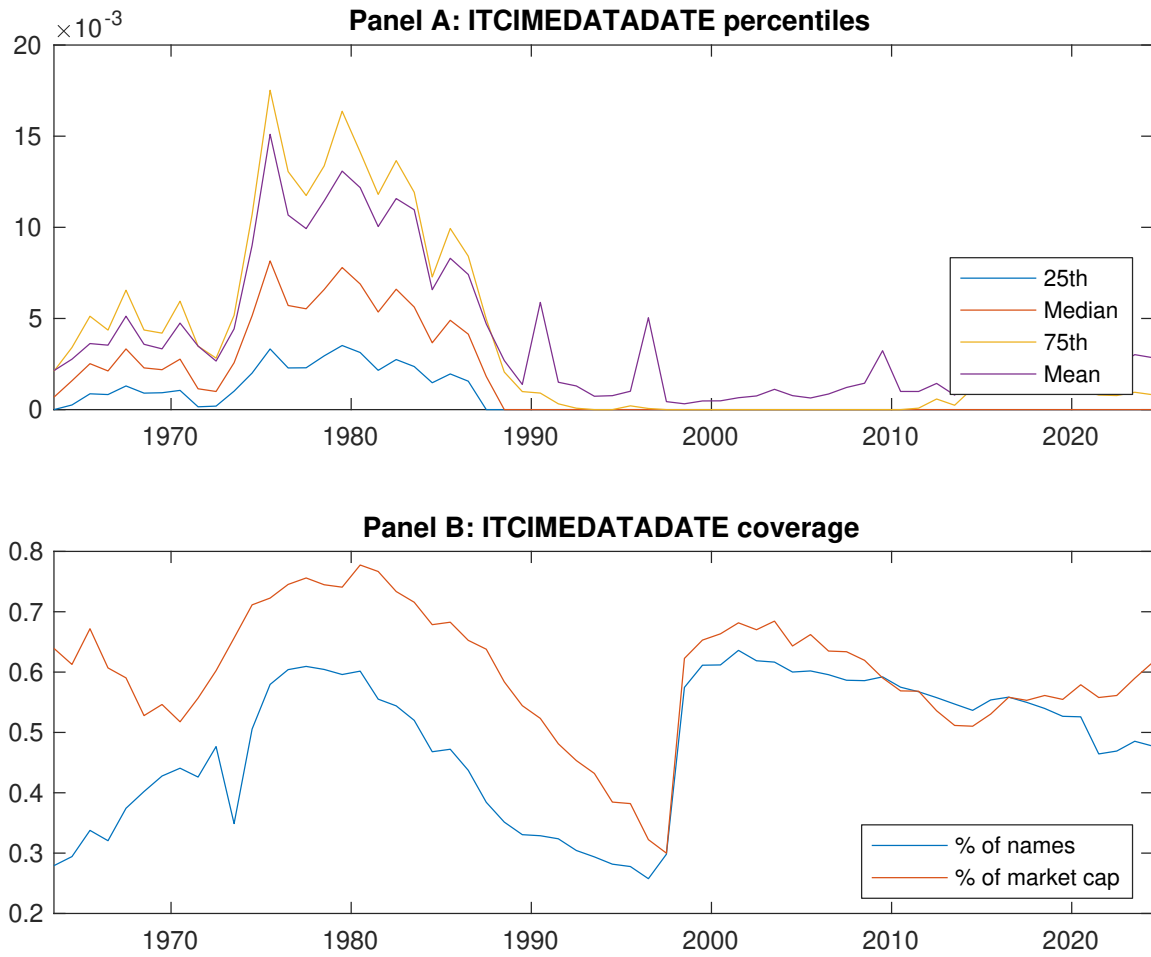


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.42]	0.57 [3.50]	0.56 [3.31]	0.73 [4.24]	0.78 [4.42]	0.36 [3.65]
α_{CAPM}	-0.17 [-3.11]	0.02 [0.38]	-0.01 [-0.11]	0.15 [2.64]	0.20 [2.96]	0.36 [3.62]
α_{FF3}	-0.16 [-3.04]	-0.01 [-0.17]	-0.05 [-0.84]	0.16 [2.76]	0.16 [2.40]	0.32 [3.21]
α_{FF4}	-0.15 [-2.72]	-0.03 [-0.56]	-0.03 [-0.55]	0.15 [2.47]	0.15 [2.24]	0.30 [2.93]
α_{FF5}	-0.25 [-4.68]	-0.08 [-1.50]	-0.08 [-1.47]	0.13 [2.22]	0.19 [2.76]	0.44 [4.31]
α_{FF6}	-0.23 [-4.25]	-0.09 [-1.70]	-0.07 [-1.16]	0.12 [2.02]	0.18 [2.61]	0.41 [3.98]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.01 [77.64]	0.98 [78.44]	0.98 [69.96]	1.00 [69.11]	1.01 [61.24]	-0.01 [-0.27]
β_{SMB}	0.04 [2.01]	0.00 [0.01]	0.00 [0.16]	-0.02 [-0.90]	-0.00 [-0.19]	-0.04 [-1.19]
β_{HML}	-0.10 [-3.95]	0.02 [0.99]	0.08 [2.97]	-0.01 [-0.50]	0.08 [2.54]	0.18 [3.79]
β_{RMW}	0.17 [6.49]	0.11 [4.46]	0.08 [2.95]	0.07 [2.52]	-0.11 [-3.56]	-0.28 [-5.80]
β_{CMA}	0.18 [4.80]	0.15 [4.21]	0.06 [1.56]	0.01 [0.22]	0.06 [1.30]	-0.12 [-1.67]
β_{UMD}	-0.03 [-2.26]	0.02 [1.30]	-0.02 [-1.70]	0.01 [1.01]	0.01 [0.65]	0.04 [1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	462	410	405	398	446	
me (\$10 ⁶)	865	933	895	1955	1374	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36 [3.65]	0.36 [3.62]	0.32 [3.21]	0.30 [2.93]	0.44 [4.31]	0.41 [3.98]
Quintile	NYSE	EW	0.35 [4.84]	0.34 [4.72]	0.35 [4.79]	0.32 [4.27]	0.42 [5.67]	0.38 [5.16]
Quintile	Name	VW	0.42 [4.26]	0.43 [4.25]	0.39 [3.90]	0.37 [3.62]	0.51 [5.09]	0.49 [4.74]
Quintile	Cap	VW	0.33 [3.30]	0.34 [3.38]	0.31 [3.09]	0.28 [2.76]	0.43 [4.24]	0.39 [3.87]
Decile	NYSE	VW	0.40 [3.38]	0.39 [3.24]	0.34 [2.85]	0.31 [2.57]	0.45 [3.70]	0.42 [3.39]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.31]	0.33 [3.28]	0.29 [2.89]	0.28 [2.75]	0.38 [3.78]	0.36 [3.66]
Quintile	NYSE	EW	0.17 [2.23]	0.16 [2.09]	0.16 [2.10]	0.15 [1.92]	0.20 [2.47]	0.18 [2.31]
Quintile	Name	VW	0.39 [3.89]	0.39 [3.91]	0.35 [3.56]	0.34 [3.42]	0.45 [4.50]	0.44 [4.39]
Quintile	Cap	VW	0.30 [3.01]	0.31 [3.08]	0.28 [2.80]	0.26 [2.61]	0.37 [3.75]	0.36 [3.58]
Decile	NYSE	VW	0.36 [3.00]	0.35 [2.91]	0.31 [2.57]	0.29 [2.42]	0.39 [3.21]	0.37 [3.09]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TSI Quintiles					TSI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.50 [2.07]	0.62 [2.56]	0.70 [2.93]	0.79 [3.32]	0.94 [3.80]	0.44 [4.25]	0.45 [4.25]	0.46 [4.41]	0.45 [4.20]	0.55 [5.14]	0.53 [4.90]
	(2)	0.66 [2.81]	0.69 [3.06]	0.73 [3.30]	0.74 [3.30]	0.99 [4.08]	0.33 [3.18]	0.34 [3.24]	0.37 [3.55]	0.32 [3.04]	0.47 [4.60]	0.43 [4.12]
	(3)	0.66 [3.17]	0.61 [2.85]	0.73 [3.47]	0.81 [3.88]	0.91 [4.23]	0.25 [2.46]	0.23 [2.26]	0.24 [2.32]	0.22 [2.12]	0.31 [3.04]	0.29 [2.84]
	(4)	0.53 [2.67]	0.66 [3.43]	0.70 [3.65]	0.71 [3.61]	0.91 [4.61]	0.38 [3.92]	0.40 [4.06]	0.39 [3.92]	0.37 [3.66]	0.47 [4.82]	0.45 [4.54]
	(5)	0.37 [2.17]	0.57 [3.47]	0.60 [3.72]	0.65 [3.72]	0.66 [3.84]	0.29 [2.45]	0.30 [2.54]	0.27 [2.24]	0.22 [1.80]	0.38 [3.16]	0.33 [2.71]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TSI Quintiles					TSI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	226	227	227	228	228	18	19	19	21	21	
	(2)	68	68	68	68	68	35	35	35	35	35	
	(3)	49	49	49	49	50	58	57	59	60	59	
	(4)	42	42	42	42	42	123	122	121	121	125	
(5)	38	38	38	38	38	839	834	1061	1200	911		

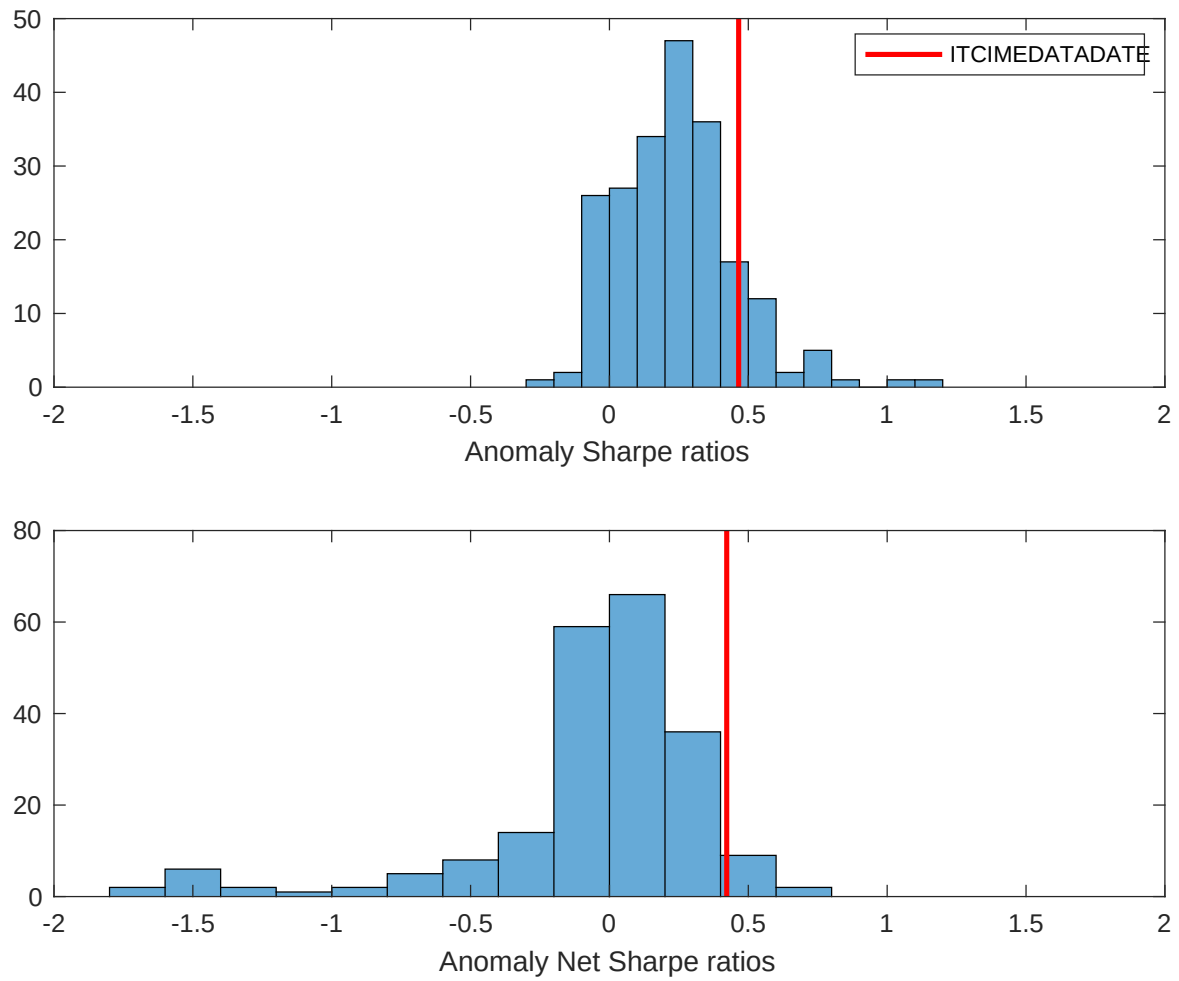


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

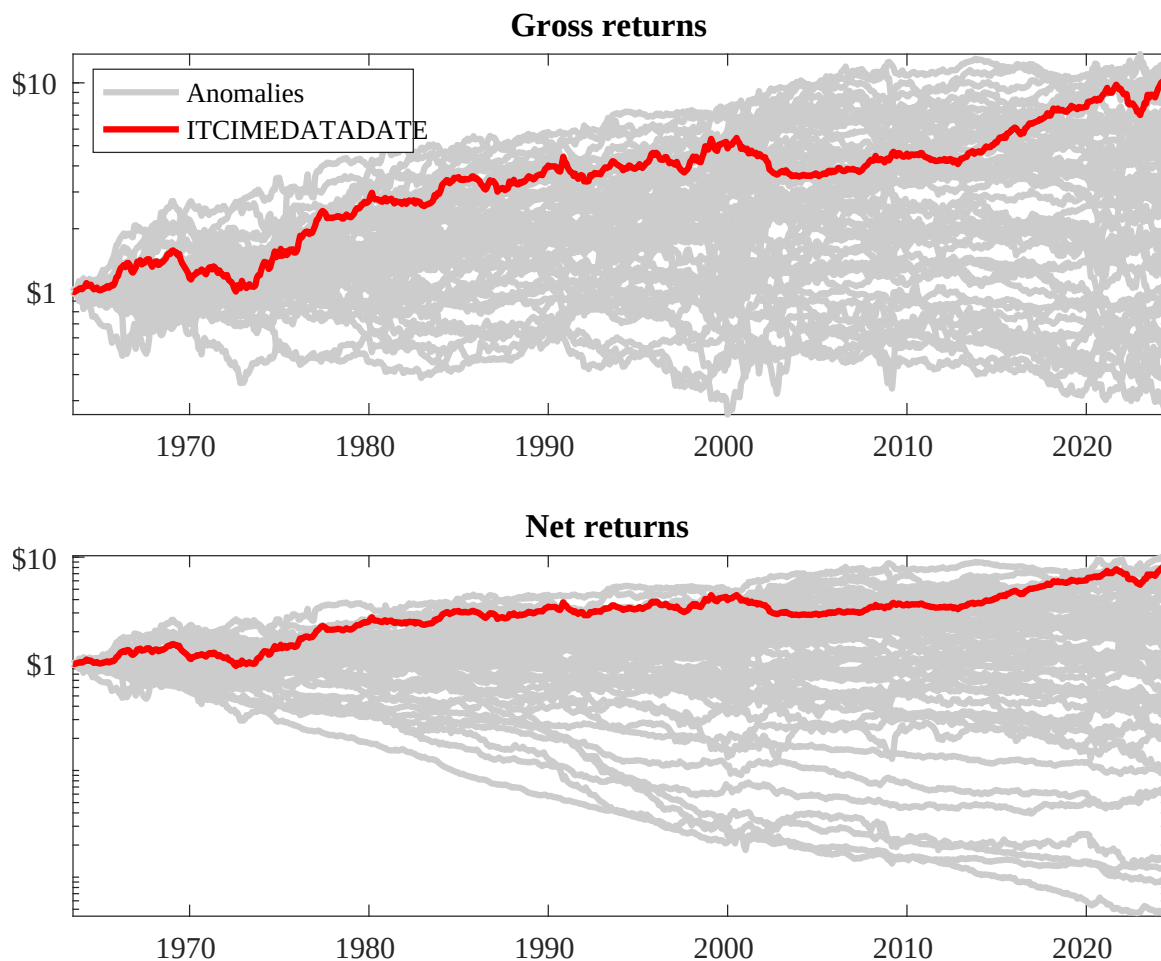
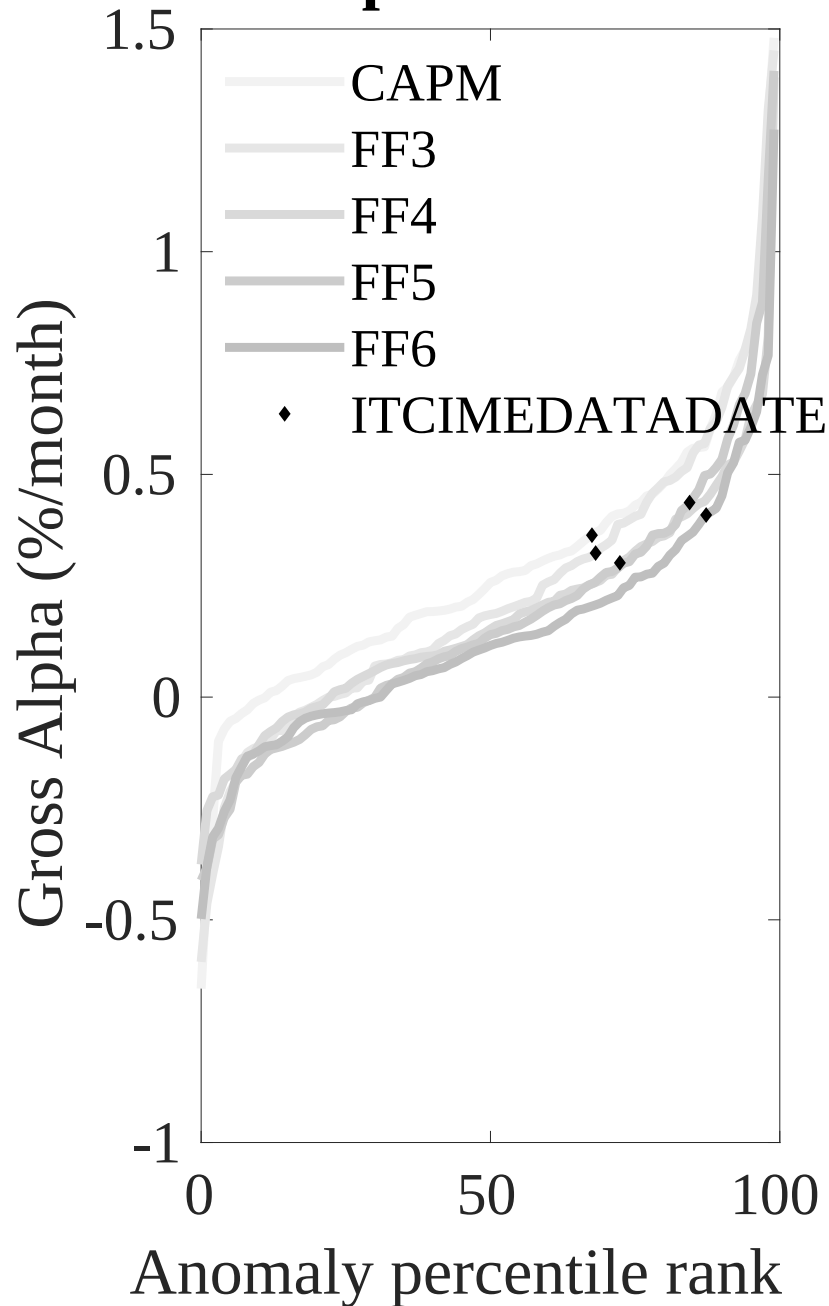


Figure 3: Dollar invested.

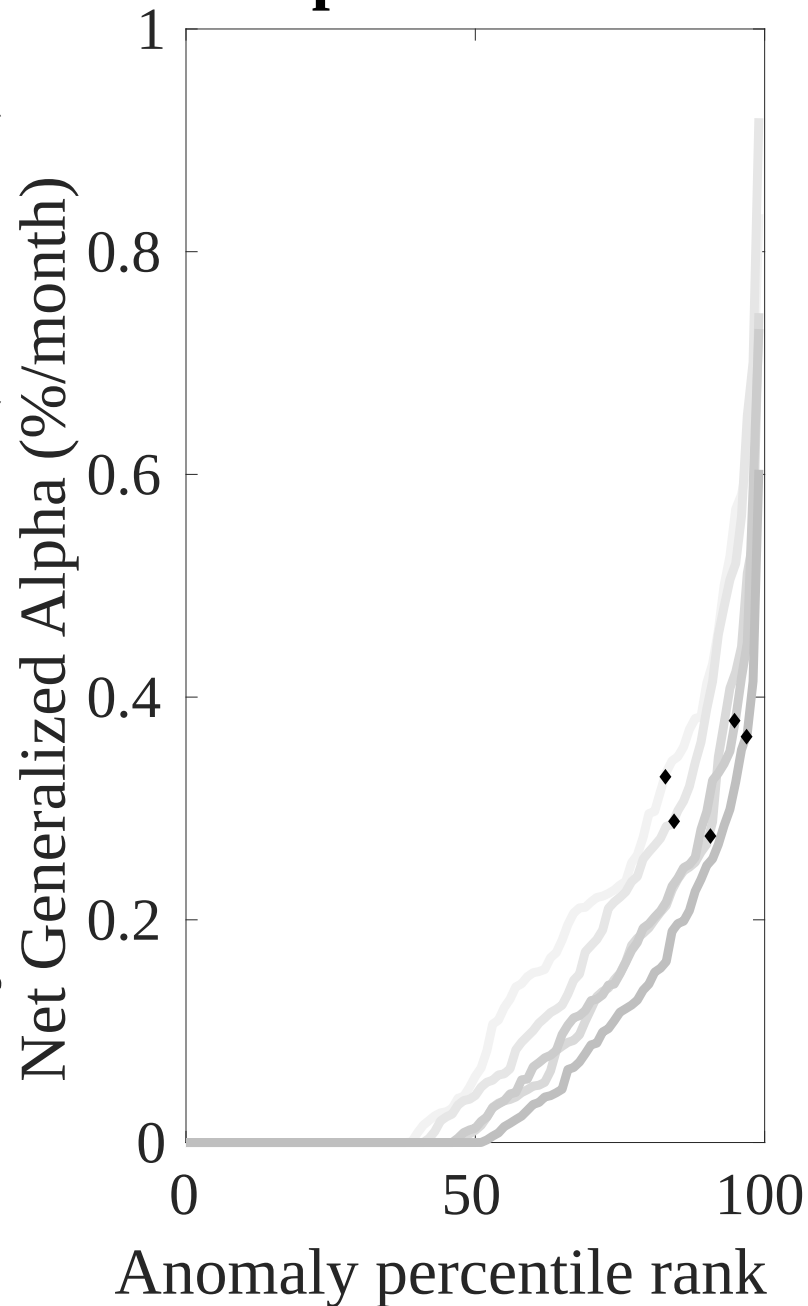
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



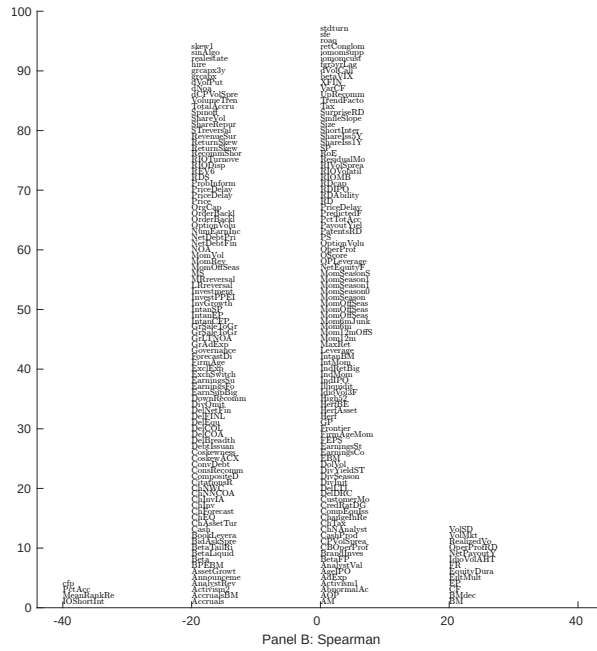
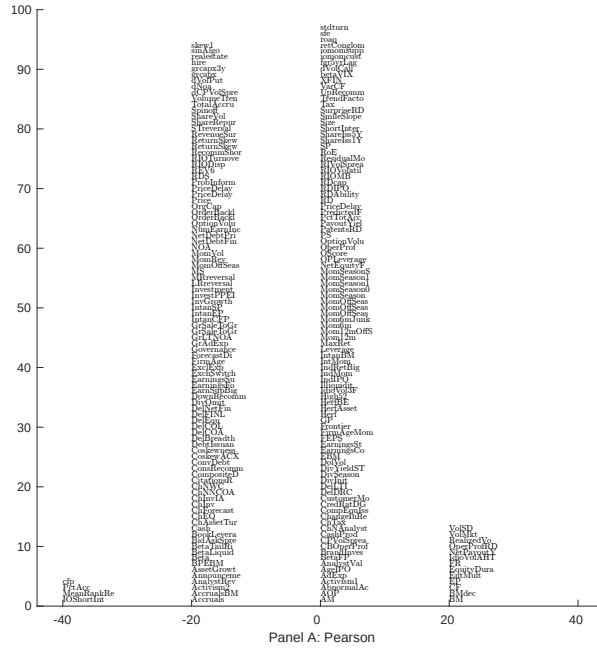


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 208 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

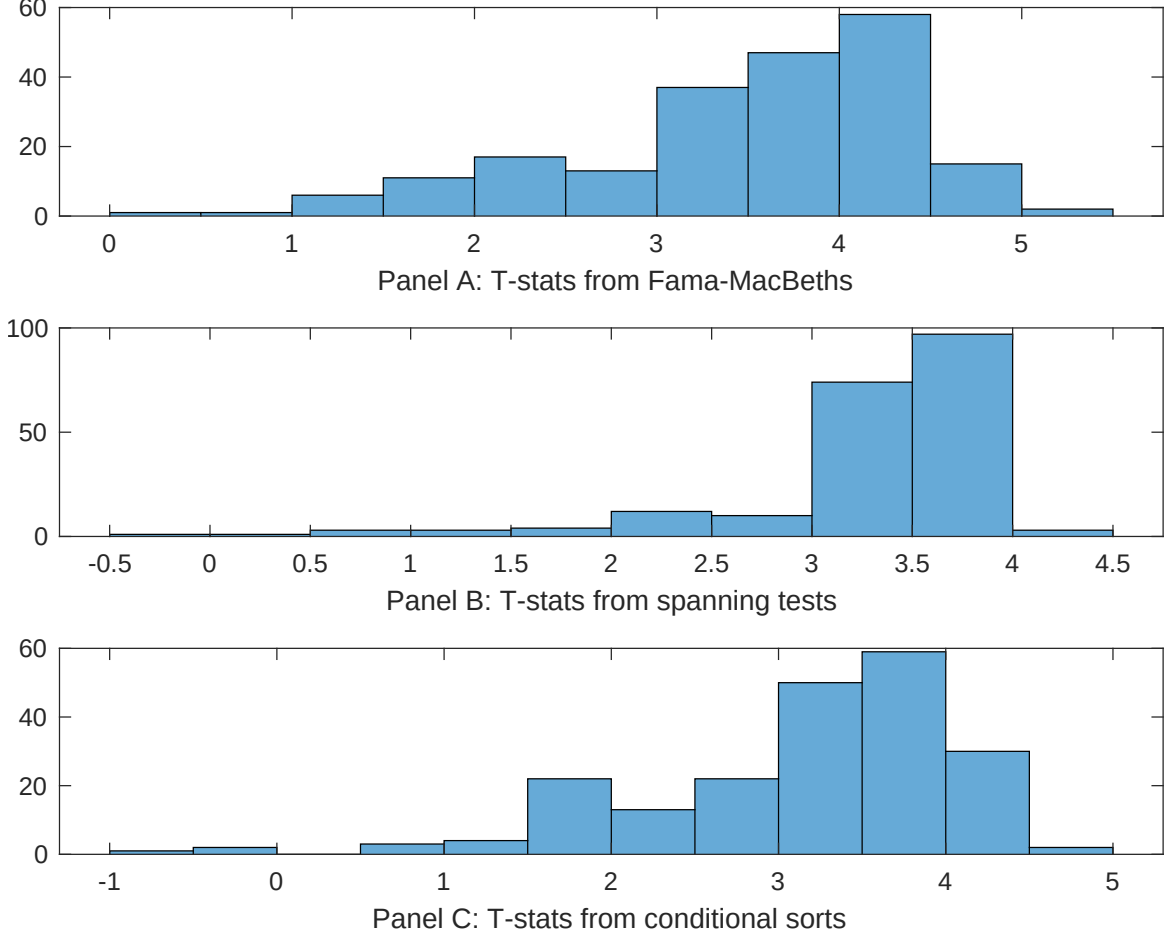


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.83 [3.71]	0.21 [9.23]	0.10 [4.70]	0.99 [4.20]	0.99 [4.42]	0.91 [3.87]	0.18 [5.21]
TSI	0.41 [3.47]	0.44 [3.68]	0.43 [3.89]	0.57 [4.55]	0.51 [4.44]	0.19 [1.32]	0.31 [1.81]
Anomaly 1	0.33 [4.95]						-0.14 [-1.39]
Anomaly 2		0.61 [5.61]					0.43 [3.16]
Anomaly 3			0.18 [2.38]				-0.30 [-2.84]
Anomaly 4				0.91 [4.72]			0.15 [4.49]
Anomaly 5					0.86 [2.84]		0.16 [1.92]
Anomaly 6						0.36 [4.90]	0.29 [3.67]
# months	727	720	727	667	727	655	631
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.44 [4.40]	0.42 [4.15]	0.38 [3.78]	0.41 [4.08]	0.34 [3.44]	0.30 [2.94]	0.34 [3.56]
Anomaly 1	42.05 [6.65]						33.56 [5.00]
Anomaly 2		18.67 [3.59]					-8.42 [-1.30]
Anomaly 3			18.79 [4.23]				5.37 [1.00]
Anomaly 4				-19.91 [-7.06]			-20.55 [-7.55]
Anomaly 5					20.17 [5.59]		17.45 [4.04]
Anomaly 6						15.62 [6.23]	10.11 [4.19]
mkt	-1.03 [-0.43]	-0.60 [-0.25]	0.30 [0.13]	-0.45 [-0.19]	-0.14 [-0.06]	-1.40 [-0.59]	-0.98 [-0.44]
smb	-16.71 [-4.14]	-8.29 [-2.18]	-6.41 [-1.78]	-5.36 [-1.55]	-5.72 [-1.63]	-6.86 [-1.96]	-20.82 [-5.38]
hml	-23.72 [-2.98]	-1.43 [-0.19]	1.62 [0.26]	30.24 [6.25]	4.86 [0.92]	21.16 [4.60]	-11.16 [-1.34]
rmw	-18.55 [-3.83]	-27.48 [-5.83]	-31.10 [-6.54]	-20.69 [-4.36]	-34.01 [-7.12]	-20.97 [-4.39]	-14.16 [-2.86]
cma	-14.36 [-2.13]	-8.91 [-1.28]	-10.24 [-1.49]	-8.36 [-1.24]	-16.91 [-2.47]	-11.77 [-1.74]	-15.58 [-2.34]
umd	3.09 [1.31]	4.50 [1.88]	7.62 [3.04]	-1.46 [-0.59]	13.52 [4.70]	8.95 [3.63]	9.08 [3.09]
# months	728	732	728	722	728	728	722
$\bar{R}^2(\%)$	11	7	8	12	10	11	22

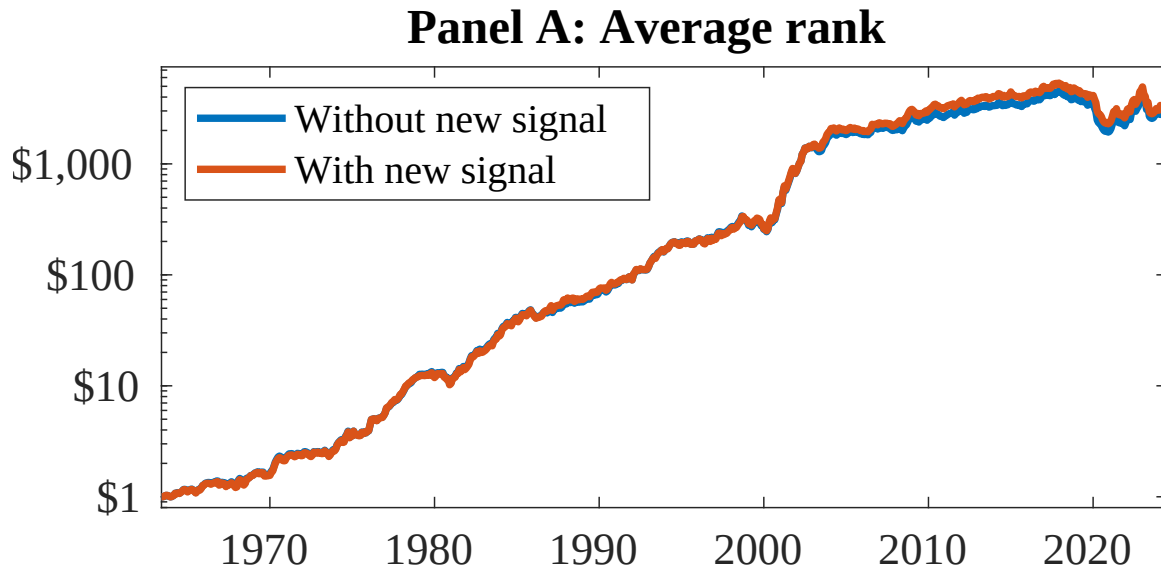


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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